Linear Algebra and Its Applications

Section 1.1: Systems of Linear Equations

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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1.1 Systems of linear equations

A solution is one ordered list of values satisfying *every* equation. The solution set is the set of all such lists. A consistent system has at least one solution; an inconsistent system has none. Two systems are equivalent when their solution sets are equal.

The three permitted operations are swapping two equations, multiplying an equation by a nonzero scalar, and adding a multiple of one equation to another. In matrix notation $R_i \leftarrow R_i + cR_j$ means to replace the *whole* row, including its last entry. We use a vertical bar to separate the coefficient columns from the constant column. A zero row means $0 = 0$; a row $[0 \ \cdots \ 0 \mid d]$ with $d \neq 0$ means an impossible equation.

Practice problem 1.1.P1: Choose the next elimination step

(a) Use the simpler coefficient to make the next pivot.

Swap equations 3 and 4, so the coefficient of x_3 in the third equation is 1 instead of 5. Next one would replace equation 4 by equation 4 minus five times equation 3. This eliminates x_3 from the last equation. Do not restart elimination on x_2: the forward phase first establishes successive pivots below the previous ones.

(b) Begin the backward phase at the last pivot.

The fourth equation is $x_4 = 1$. Add twice equation 4 to equation 1 to remove its term $-2x_4$. The other two equations already have no x_4 term. After that, divide equation 3 by 2 and use it to clear the x_3 terms above it. Each operation preserves all solutions.

Practice problem 1.1.P2: Consistency of a triangular system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 5 & 2 & -6 \\ 0 & 4 & -7 & 2 \\ 0 & 0 & 5 & 0 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{4})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 2 & -6 \\ 0 & 1 & -\frac{7}{4} & \frac{1}{2} \\ 0 & 0 & 5 & 0 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 2 & -6 \\ 0 & 1 & -\frac{7}{4} & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 0 & -6 \\ 0 & 1 & -\frac{7}{4} & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{7}{4})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 0 & -6 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (5)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{17}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -\frac{17}{2} \\ x_2 = \frac{1}{2} \\ x_3 = 0 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Practice problem 1.1.P3: Test a proposed solution in every equation

Substitute $(x_1, x_2, x_3) = (3, 4, -2)$ into the three left sides:

$$5(3) - 4 + 2(-2) = 7, \qquad -2(3) + 6(4) + 9(-2) = 0,$$

$$-7(3) + 5(4) - 3(-2) = 5.$$

The first two constants match, but the third required constant is -7, not 5. One failed equation is enough to reject the proposed solution.

Conclusion. $(3, 4, -2)$ is not a solution.

Practice problem 1.1.P4: A consistency condition on two constants

The second left side is -3 times the first. Replace equation 2 by equation 2 plus three times equation 1. The resulting last equation is $0 = k + 3h$. It can hold precisely when $k = -3h$. In that case choose any $x_2 = t$ and put $x_1 = (h + t)/2$; these values satisfy both equations.

Conclusion. Consistency holds if and only if $k = -3h$.

Exercise 1.1.1: A two-equation elimination

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ -2 & -7 & -5 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & 3 & 9 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (5)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -8 \\ 0 & 1 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -8 \\ x_2 = 3 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.2: A second two-equation elimination

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 3 & 6 & -3 \\ 5 & 7 & 10 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{3})R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 5 & 7 & 10 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (5)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & -3 & 15 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 1 & -5 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 9 \\ 0 & 1 & -5 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 9 \\ x_2 = -5 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.3: The intersection of two lines

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 1 & -1 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & -3 & -3 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 2 \\ x_2 = 1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.4: A second line intersection

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 2 & -13 \\ 3 & -2 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -13 \\ 0 & -8 & 40 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{8})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -13 \\ 0 & 1 & -5 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -3 \\ 0 & 1 & -5 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -3 \\ x_2 = -5 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.5: Begin backward elimination

The last two equations already give $x_3 = 2$ and $x_4 = -5$. To begin clearing above the x_3 pivot, subtract four times row 3 from row 2, then add three times row 3 to row 1:

$$R_2 \leftarrow R_2 - 4R_3, \qquad R_1 \leftarrow R_1 + 3R_3.$$

The two changed equations become $x_2 = -2$ and $x_1 - 4x_2 = 13$. Both operations act on all entries, including the constants.

Conclusion. Subtract four times row 3 from row 2; add three times row 3 to row 1.

Exercise 1.1.6: Create the last pivot

Use the coefficient 1 in row 3 to remove the coefficient 4 below it:

$$R_4 \leftarrow R_4 - 4R_3 \quad \Longrightarrow \quad [0\ 0\ 0\ -7 \mid 14].$$

Now multiply the last row by $-1/7$, giving $[0\ 0\ 0\ 1 \mid -2]$. These steps create the next pivot without disturbing the earlier zeros.

Conclusion. Subtract four times row 3 from row 4, then divide row 4 by -7.

Exercise 1.1.7: A contradiction row cannot be repaired

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & -4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_3 \leftrightarrow R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & -4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-4)R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & 0 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-2)R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (7)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.1.8: Four variables fixed at zero

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{3})R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{array}\right].$$

Use $R_4 \leftarrow (\frac{1}{2})R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-5)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \\ x_4 = 0 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.9: Back substitution through four pivots

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -5 \\ 0 & 1 & -2 & 0 & -7 \\ 0 & 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-3)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -5 \\ 0 & 1 & -2 & 0 & -7 \\ 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 0 & 21 \\ 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 16 \\ 0 & 1 & 0 & 0 & 21 \\ 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 16 \\ x_2 = 21 \\ x_3 = 14 \\ x_4 = 4 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.10: A triangular four-variable system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & -2 & -7 \\ 0 & 1 & 0 & 3 & 6 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-2)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & -11 \\ 0 & 1 & 0 & 3 & 6 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & -11 \\ 0 & 1 & 0 & 0 & 12 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -47 \\ 0 & 1 & 0 & 0 & 12 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -47 \\ x_2 = 12 \\ x_3 = 2 \\ x_4 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.11: A system with a zero leading coefficient

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 0 & 1 & 5 & -4 \\ 1 & 4 & 3 & -2 \\ 2 & 7 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftrightarrow R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 2 & 7 & 1 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & -1 & -5 & 2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-4)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -17 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.1.12: A three-variable contradiction

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 2 & -7 & 3 & -2 \\ -2 & 1 & 7 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 3 & -5 & 4 \\ -2 & 1 & 7 & -1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-2)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 3 & -5 & 4 \\ 0 & -9 & 15 & -7 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & -9 & 15 & -7 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-9)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & 0 & 0 & 5 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & 0 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (\frac{4}{3})R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & 0 \\ 0 & 1 & -\frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-5)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & -\frac{13}{3} & 0 \\ 0 & 1 & -\frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.1.13: A unique three-variable solution

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 2 & 2 & 9 & 7 \\ 0 & 1 & 5 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 2 & 15 & -9 \\ 0 & 1 & 5 & -2 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{2})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 1 & 5 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 0 & -\frac{5}{2} & \frac{5}{2} \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{2}{5})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (\frac{15}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 5 \\ x_2 = 3 \\ x_3 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.14: Solve after scaling a leading row

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 2 & 0 & -6 & -8 \\ 0 & 1 & 2 & 3 \\ 3 & 6 & -2 & -4 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{2})R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 3 & 6 & -2 & -4 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 6 & 7 & 8 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (6)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -5 & -10 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 2 \\ x_2 = -1 \\ x_3 = 2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.15: Detect inconsistency without solving

The augmented rows are

$$\begin{bmatrix} 1 & -6 & 0 & 0 & 5 \\ 0 & 1 & -4 & 1 & 0 \\ -1 & 6 & 1 & 5 & 3 \\ 0 & -1 & 5 & 4 & 0 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 + R_1$ and $R_4 \leftarrow R_4 + R_2$. Rows 3 and 4 become $[0\ 0\ 1\ 5 \mid 8]$ and $[0\ 0\ 1\ 5 \mid 0]$. Then $R_4 \leftarrow R_4 - R_3$ gives $[0\ 0\ 0\ 0 \mid -8]$. Its equation $0 = -8$ is impossible.

Conclusion. The system is inconsistent. No further solving is needed.

Exercise 1.1.16: Detect consistency without solving

Divide row 1 by 2, then use $R_4 \leftarrow R_4 + 3R_1$:

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -5 \\ 0 & 3 & 3 & 0 & 0 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 2 & 3 & -5 & -10 \end{bmatrix}.$$

Now $R_4 \leftarrow R_4 - \frac{2}{3}R_2$ gives $[0\ 0\ 1\ -5 \mid -10]$, and $R_4 \leftarrow R_4 - R_3$ gives $[0\ 0\ 0\ -9 \mid -9]$. There is a nonzero pivot in each variable column and no contradiction row. Back substitution is therefore possible from the last equation upward.

Conclusion. The system is consistent. The exercise does not require its values.

Exercise 1.1.17: Do three lines meet at one point?

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 2 & 3 & -1 \\ 6 & 5 & 0 \\ 2 & -5 & 7 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{2})R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 6 & 5 & 0 \\ 2 & -5 & 7 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (6)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -4 & 3 \\ 2 & -5 & 7 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -4 & 3 \\ 0 & -8 & 8 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{4})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{3}{4} \\ 0 & -8 & 8 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-8)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 2 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{3}{4})R_3$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (\frac{3}{2})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.1.18: Do three planes have a common point?

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 2 & 4 & 4 & 4 \\ 0 & 1 & -2 & -2 \\ 2 & 3 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{2})R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 2 & 3 & 0 & 0 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 0 & -1 & -4 & -4 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & -6 & -6 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{6})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.19: A forbidden value of the parameter

Replace row 2 by row 2 minus three times row 1:

$$\begin{bmatrix} 1 & h & 4 \\ 0 & 6-3h & -4 \end{bmatrix}.$$

If $h = 2$, the last equation is $0 = -4$, so there is no solution. If $h \neq 2$, the second variable is $-4/(6-3h)$ and the first is then determined by row 1. Thus a solution does exist in every other case.

Conclusion. $h \neq 2$.

Exercise 1.1.20: A second forbidden parameter value

Use $R_2 \leftarrow R_2 - 2R_1$ to obtain $[0\ -8-2h \mid 16]$. If $h = -4$, this asserts $0 = 16$. For $h \neq -4$ the nonzero coefficient determines x_2, then row 1 determines x_1.

Conclusion. Consistency holds precisely for $h \neq -4$.

Exercise 1.1.21: A parameter that never destroys consistency

After $R_2 \leftarrow R_2 - 3R_1$, the last row is $[0\ h-12 \mid 0]$. If $h \neq 12$ it says $x_2 = 0$ and then $x_1 = -2$. If $h = 12$ it says $0 = 0$, leaving x_2 free and $x_1 = -2 - 4x_2$. Neither case produces a contradiction.

Conclusion. Every real h gives a consistent system.

Exercise 1.1.22: Match a multiple equation

Use $R_1 \leftarrow R_1 + 2R_2$. Its result is $[0\ 0 \mid h-6]$. Thus consistency requires $h = 6$. Conversely, if $h = 6$, the first equation is -2 times the second, so solving $2x_1 - 6x_2 = -3$ supplies solutions of both.

Conclusion. $h = 6$.

Exercise 1.1.23: Definitions and reversible operations

(a) True. Swapping twice restores the original rows. Scaling by $c \neq 0$ is reversed by scaling by $1/c$. Adding cR_j to R_i is reversed by adding $-cR_j$ to R_i.

(b) False. In an $m \times n$ matrix, m counts rows and n counts columns. A 5×6 matrix has five rows and six columns.

(c) False. The described list is *one solution*. The solution *set* contains all lists satisfying the system, and may be empty, a singleton, or infinite.

(d) True. Existence asks whether at least one solution exists; uniqueness asks whether exactly one exists. These are separate questions.

Exercise 1.1.24: Row equivalence and consistency

(a) False. Equal row counts do not imply row equivalence. For example, [0 0] cannot be row reduced to [1 0]: all allowed operations keep a zero row zero.

(b) True. Each row operation replaces equations by reversible consequences, so it preserves the common solution set.

(c) False. Having identical solution sets is the definition of equivalent systems; different sets would contradict that definition.

(d) True. "Consistent" means that at least one solution exists.

Exercise 1.1.25: Compatibility of dependent rows

The coefficient rows satisfy $r_3 = -2r_1 - r_2$. Make this relation explicit by $R_3 \leftarrow R_3 + 2R_1$ and then $R_3 \leftarrow R_3 + R_2$. The augmented last row becomes $[0\ 0\ 0 \mid k + 2g + h]$. Therefore consistency requires $k + 2g + h = 0$. If it holds, the remaining first two rows have nonzero pivots in columns 1 and 2, so they can be solved with x_3 free.

Conclusion. $2g + h + k = 0$ is necessary and sufficient.

Exercise 1.1.26: Consistency for every right side

Replace row 2 by row 2 minus $c/2$ times row 1. This gives

$$[0\ d - 2c \mid g - cf/2].$$

If $d = 2c$, choose $f = 0, g = 1$. The resulting row asserts $0 = 1$, contradicting the assumption of consistency for *every* right side. Thus $d \neq 2c$. Conversely, that inequality supplies a second pivot for any f, g, so it also suffices.

Conclusion. $d \neq 2c$.

Exercise 1.1.27: The condition on a general two-by-two system

Because $a \neq 0$, use $R_2 \leftarrow R_2 - (c/a)R_1$. The new row is

$$[0\ d - cb/a \mid g - cf/a].$$

Its coefficient cannot be zero: otherwise $f = 0, g = 1$ would make the system inconsistent. Multiplying $d - cb/a \neq 0$ by the nonzero a gives $ad - bc \neq 0$. Conversely, this condition makes the second pivot nonzero for every right side.

Conclusion. $ad - bc \neq 0$. The condition states exactly that the two equations have independent left sides.

Exercise 1.1.28: Construct three equivalent augmented matrices

Start with the equations $x_1 = 3$, $x_2 = -2$, $x_3 = -1$. Three possible augmented matrices are

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -1 \end{array}\right], \qquad \left[\begin{array}{ccc|c} 2 & 0 & 0 & 6 \\ 0 & 3 & 0 & -6 \\ 0 & 0 & 4 & -4 \end{array}\right],$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

The second merely rescales equations by nonzero numbers. The third comes from adding row 2 to row 1 and then the original row 3 to row 2. Alternatively back substitution in the third gives $x_3 = -1$, $x_2 = -2$, $x_1 = 3$. Thus all three have exactly the requested single solution.

Exercise 1.1.29: Reverse a row interchange

The first and third rows exchange places; the middle row stays fixed.

$$\text{Forward: } R_1 \leftrightarrow R_3, \qquad \text{reverse: } R_1 \leftrightarrow R_3.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

Exercise 1.1.30: Reverse a row scaling

Each entry in $[0, -5, 10]$ is multiplied by $-1/5$, giving $[0, 1, -2]$.

$$\text{Forward: } R_3 \leftarrow -\tfrac{1}{5}R_3, \qquad \text{reverse: } R_3 \leftarrow -5R_3.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

Exercise 1.1.31: Reverse a multiple-row subtraction

The subtraction is $[4, -1, 3, -6] - 4[1, -2, 1, 0] = [0, 7, -1, -6]$.

$$\text{Forward: } R_3 \leftarrow R_3 - 4R_1, \qquad \text{reverse: } R_3 \leftarrow R_3 + 4R_1.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

Exercise 1.1.32: Reverse a second multiple-row subtraction

The subtraction is $[0, 4, -12, 7] - 4[0, 1, -3, -2] = [0, 0, 0, 15]$.

$$\text{Forward: } R_3 \leftarrow R_3 - 4R_2, \qquad \text{reverse: } R_3 \leftarrow R_3 + 4R_2.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

Exercise 1.1.33: Set up the four-node temperature model

At each node, multiply "temperature equals the average of its four neighbors" by 4. The two fixed neighboring boundary values contribute to the constant; the two interior neighbors stay as unknowns. Reading the labels in the figure gives

$$\begin{aligned} 4T_1 - T_2 - T_4 &= 10 + 20 = 30, \\ -T_1 + 4T_2 - T_3 &= 20 + 40 = 60, \\ -T_2 + 4T_3 - T_4 &= 40 + 30 = 70, \\ -T_1 - T_3 + 4T_4 &= 10 + 30 = 40. \end{aligned}$$

Conclusion. These four equations are the required equilibrium model. The order of nodes around the square is $1, 2, 3, 4$, not a row-wise ordering.

Exercise 1.1.34: Solve the temperature model

Use the equations from Exercise 33. Adding the first and third gives $4(T_1 + T_3) - 2(T_2 + T_4) = 100$. Adding the second and fourth gives $-2(T_1 + T_3) + 4(T_2 + T_4) = 100$. Let $S = T_1 + T_3$ and $U = T_2 + T_4$. Dividing both equations by 2 gives $2S - U = 50$ and $-S + 2U = 50$. Thus $S = U = 50$. Now substitute these pair sums into the original equations:

$$T_1 = (30 + U)/4 = 20, \quad T_2 = (60 + S)/4 = 55/2,$$

$$T_3 = (70 + U)/4 = 30, \quad T_4 = (40 + S)/4 = 45/2.$$

For example, the first average is $(10 + 20 + 55/2 + 45/2)/4 = 20$; the other three check in the same way by the four displayed equations. Every solution must have these pair sums and hence these values, so uniqueness is also established.

Conclusion. $T_1 = 20$, $T_2 = 27.5$, $T_3 = 30$, $T_4 = 22.5$ degrees.